EE2021 - Materials Science for Electrical Engineers

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Technology

- The knowledge and devices that humans use to accomplish the purposes of life and master the natural environment (eg: spectacles, mechanical clock, printing press)
- Artifacts not readily available in nature but created by means of complex processing (material, information, energy)
- Is Technology an applied science? - Material transformation in past has come from craftsmen's observation without any theory or philosophy (paper, cast iron, gun powder, glass)
- Many times in history, technology even preceded science and sometimes aided in the further progress of science
- Art of making approximations (constraints) and decisions on incomplete data
- Solutions could be double edged and morally ambiguous

R. Friedel, A Culture of Improvement: Technology and the Western Millenium, MIT Press, 2007
D. S. L. Cardwell, *Turning points in Western technology*, 1991

Activity- Guess how many elements are needed to create an iPhone?

iPhone 6, 16GB model

Element	Chemical Symbol	Percent of iPhone by weight	Grams used in iPhone	Average cost per gram	Value of element in iPhone
Aluminum	Al	24.14	31.14	$ 0.0018	$ 0.055
Arsenic	As	0.00	0.01	$ 0.0022	$-
Gold	Au	0.01	0.014	$ 40.00	$ 0.56
Bismuth	Bi	0.02	0.02	$ 0.0110	$ 0.0002
Carbon	C	15.39	19.85	$ 0.0022	$-
Calcium	Ca	0.34	0.44	$ 0.0044	$ 0.002
Chlorine	Cl	0.01	0.01	$ 0.0011	$-
Cobalt	Co	5.11	6.59	$ 0.0396	$ 0.261
Chrome	Cr	3.83	4.94	$ 0.0020	$ 0.010
Copper	Cu	6.08	7.84	$ 0.0059	$ 0.047
Iron	Fe	14.44	18.63	$ 0.0001	$ 0.002
Gallium	Ga	0.01	0.01	$ 0.3304	$ 0.003
Hydrogen	H	4.28	5.52	$-	$-
Potassium	K	0.25	0.33	$ 0.0003	$-
Lithium	Li	0.67	0.87	$ 0.0198	$ 0.017
Magnesium	Mg	0.51	0.65	$ 0.0099	$ 0.006
Manganese	Mn	0.23	0.29	$ 0.0077	$ 0.002
Molybdenum	Mo	0.02	0.02	$ 0.0176	$ 0.000
Nickel	Ni	2.10	2.72	$ 0.0099	$ 0.027
Oxygen	O	14.50	18.71	$-	$-
Phosphorus	P	0.03	0.03	$ 0.0001	$-
Lead	Pb	0.03	0.04	$ 0.0020	$-
Sulfur	S	0.34	0.44	$ 0.0001	$-
Silicon	Si	6.31	8.14	$ 0.0001	$ 0.001
Tin	Sn	0.51	0.66	$ 0.0198	$ 0.013
Tantalum	Ta	0.02	0.02	$ 0.1322	$ 0.003
Titanium	Ti	0.23	0.30	$ 0.0198	$ 0.006
Tungsten	W	0.02	0.02	$ 0.2203	$ 0.004
Vanadium	V	0.03	0.04	$ 0.0991	$ 0.004
Zinc	Zn	0.54	0.69	$ 0.0028	$ 0.002
	TOTAL	100%	129 grams		$ 1.03

Equipment Used



PHOTO 1 – WEIGHT DROPPER



PHOTO 2 – BLENDER

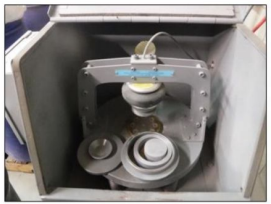
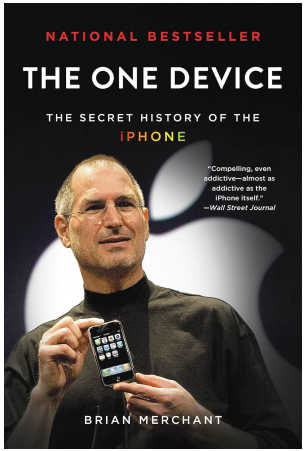


PHOTO 3 – RING AND PUCK PULVERIZER



https://www.vice.com/en/article/everything-thats-inside-your-iphone/

Material vs Matter

"*Materials themselves have interacted with mainstream history, for they are the stuff on which virtually all human activities are based*" - Cyril Stanley Smith

"The Age of Stone, or that period when weapons and implements were made of stone, wood, bone, or some such material, and during which very little or nothing at all was known of metals....

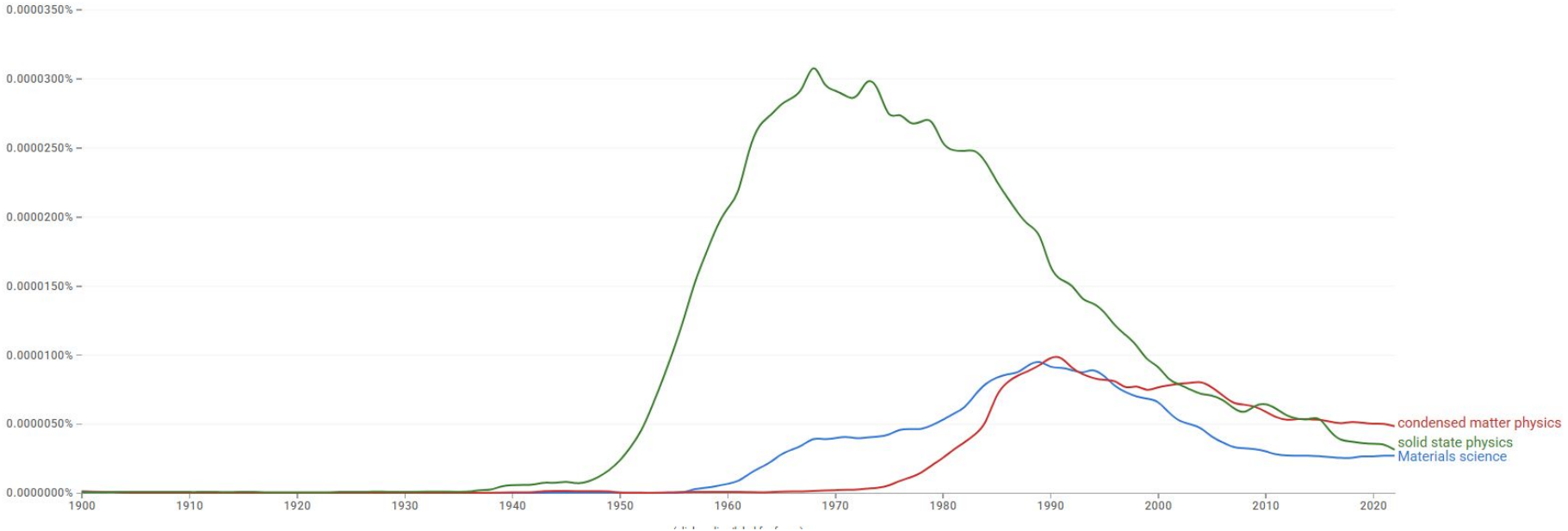
"The Age of Bronze, in which weapons and cutting implements were made of copper or bronze, and nothing at all, or but very little was known of iron or silver....

"The Age of Iron is the third and last period of the heathen times, in which iron was used for those articles to which that metal is eminently suited, and in the fabrication of which it came to be employed as a substitute for bronze" (Christian Thomsen, [1848]).

Materials - substances having properties which make them useful in machines, structures, devices, and products[#].

[#] *Materials and Man's Needs* - COSMAT (Committee on the Survey of Materials Science and Engineering) Report, 1970

What is in a name?



https://books.google.com/ngrams/graph?content=Materials+science%2C+condensed+matter+physics%2C+solid+state+physics&year_start=1900&year_end=2022&corpus=en&smoothing=3

What is in a name?

Condensed matter physics is the largest sub-field of physics. It deals with the macroscopic and microscopic physical properties of the "condensed" phases of matter

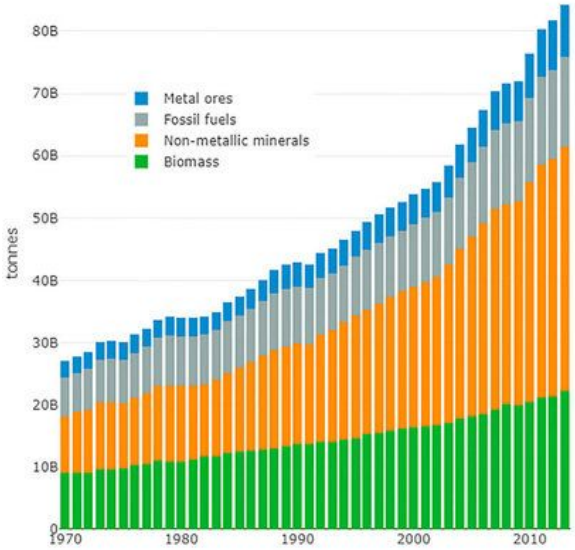
Solid state physics is the biggest sub-field of condensed matter physics. It is less glamorous than High Energy Physics or Cosmology but economically and technologically more relevant than any subfield of Physics[2,3]

Materials science and engineering is concerned with the generation and application of knowledge relating the composition, structure, and processing of materials to their properties and performance.[1,4] It is a blend of metallurgy, ceramics/polymer engg, condensed matter physics and physical chemistry.

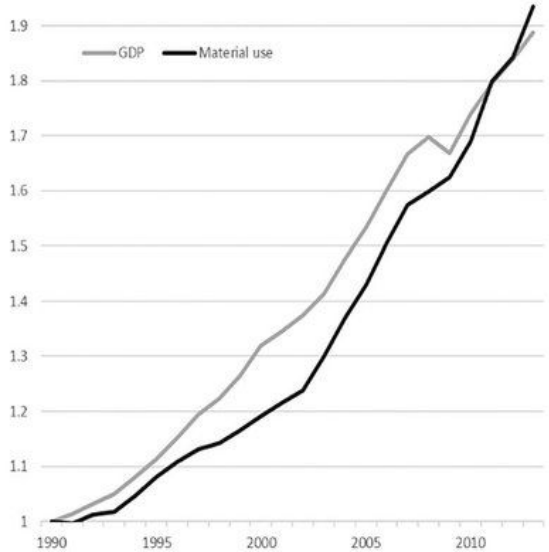
1. "*Materials Science*" by Cyrus C. M. Mody and Joseph D. Martin
2. *What's in a Name Change? - Solid State Physics, Condensed Matter Physics, and Materials Science* by Joseph D. Martin
3. Joseph Martin, Solid State Insurrection: How the Science of Substance Made American Physics Matter, UPP 2018
4. *Materials and Man's Needs* - COSMAT (Committee on the Survey of Materials Science and Engineering) Report, 1970

Materials consumption (2020)

Global material footprint 1970-2013



Global GDP and material footprint 1990-2013



Concrete - 38 Gt

Cement - 4.4 Gt

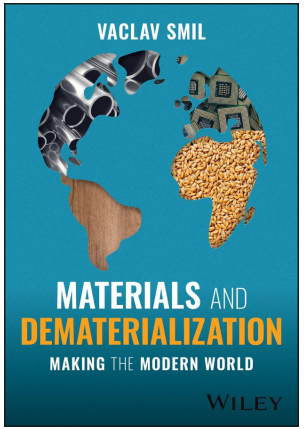
Steel - 1.88 Gt

Wood - 830 Mt

Paper- 400 Mt

Plastic - 370 Mt

Aluminium- 65 Mt



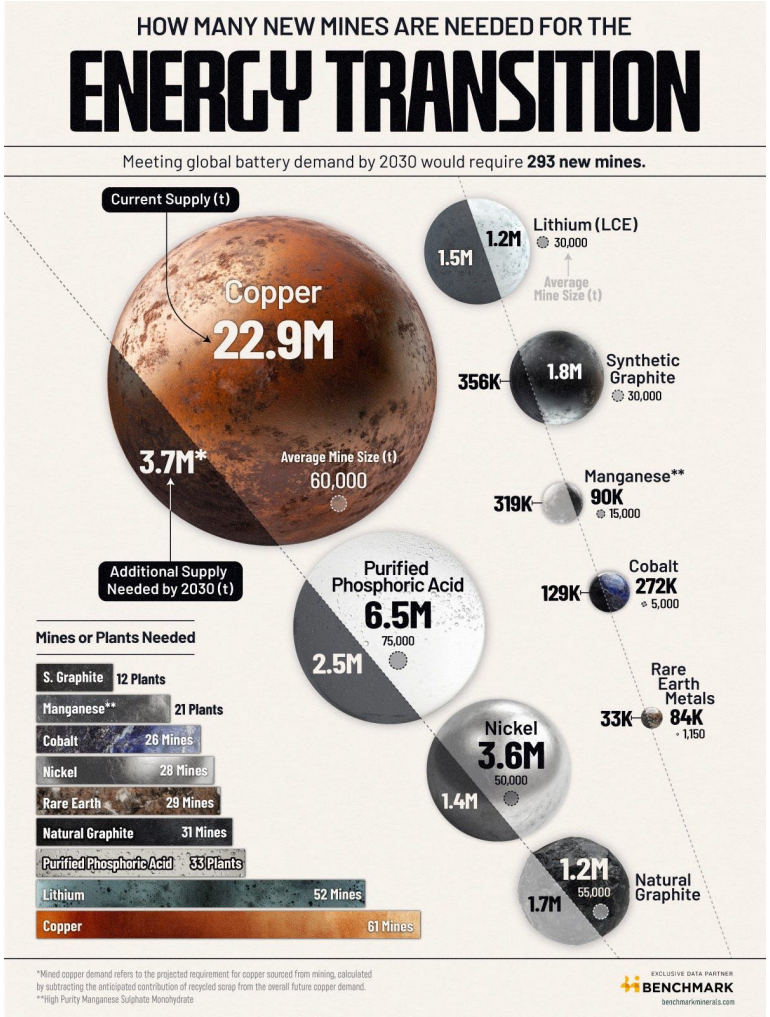
How far we can push relative dematerialization?

*We're also challenging ourselves to **one day end our reliance on mining** altogether - Apple (2017)*

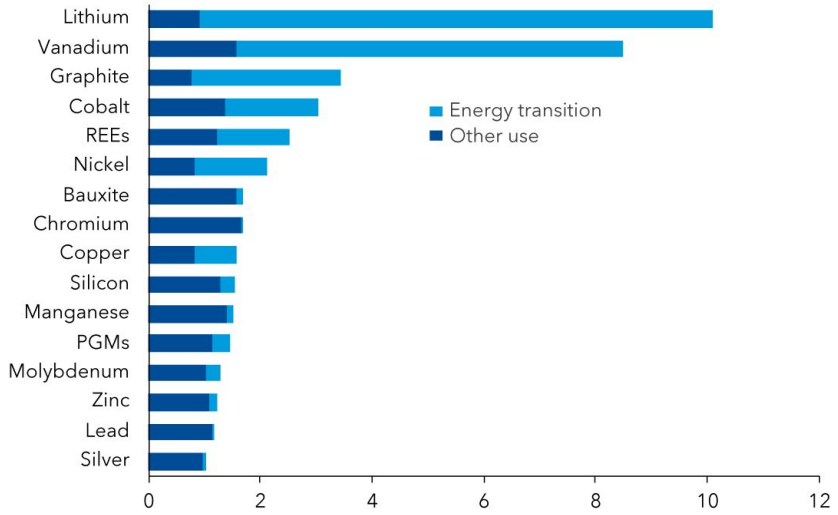
Is an economy possible where circular loops of material, energy and waste flows encompass all societal activities?

Hickel, Jason & Kallis, Giorgos. (2019). Is Green Growth Possible?. New Political Economy. 25. 1-18
https://images.apple.com/environment/pdf/Apple_Environmental_Responsibility_Report_2017.pdf

Clean Energy is a Dirty Business



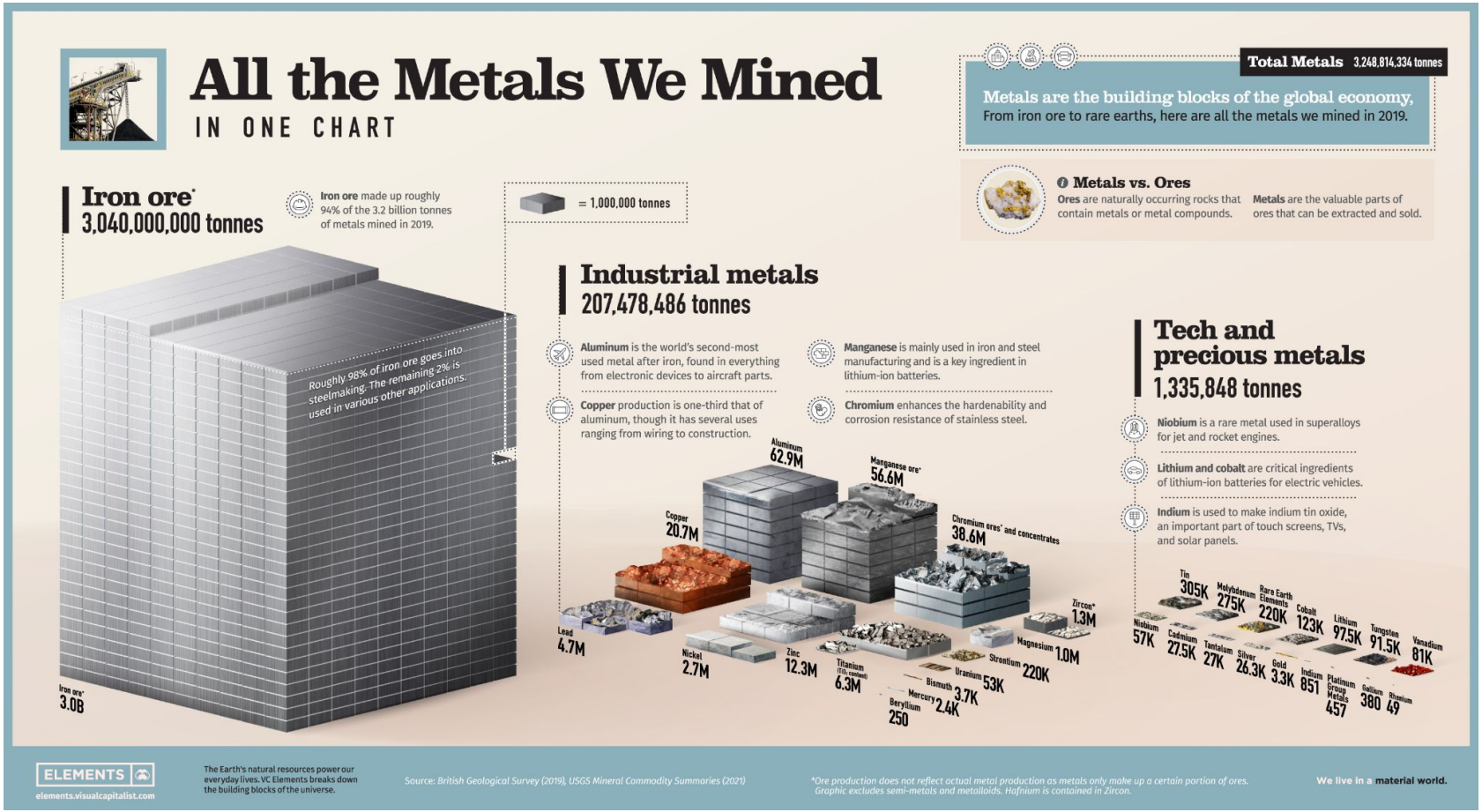
Ratio of 2050 to 2022 demand under a net zero emissions scenario



Sources: International Energy Agency (IEA) World Energy Outlook (2023); and IMF staff calculations.
Note: The chart shows the IEA's projected increase in mineral demand (in quantity terms) broken down by sector as a ratio of 2050 to 2022 demand, under the IEA's net zero emissions transition scenario. REE = Rare Earth Elements; PGMs = Platinum Group Metals.

IMF

More mining and materials are needed in the future



Why is iron mined more than other elements?

Three likely answers?

https://elements.visualcapitalist.com/wp-content/uploads/2021/09/all-of-the-metals-one-visualization.html

Relative abundance of elements in Solar System

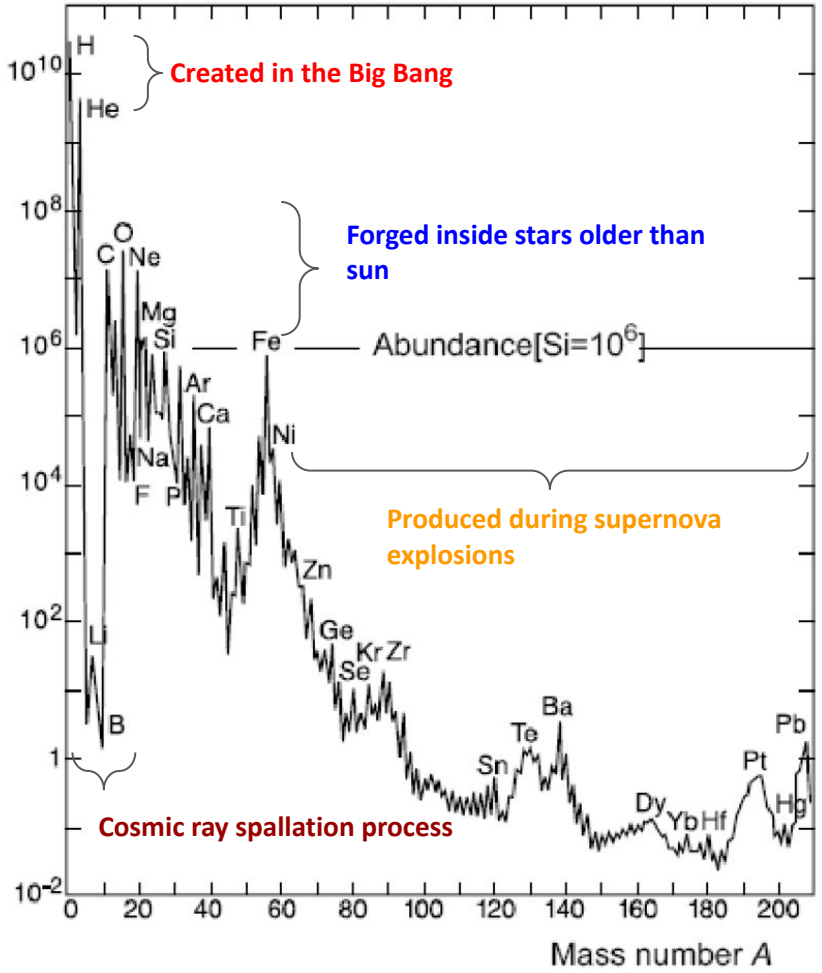


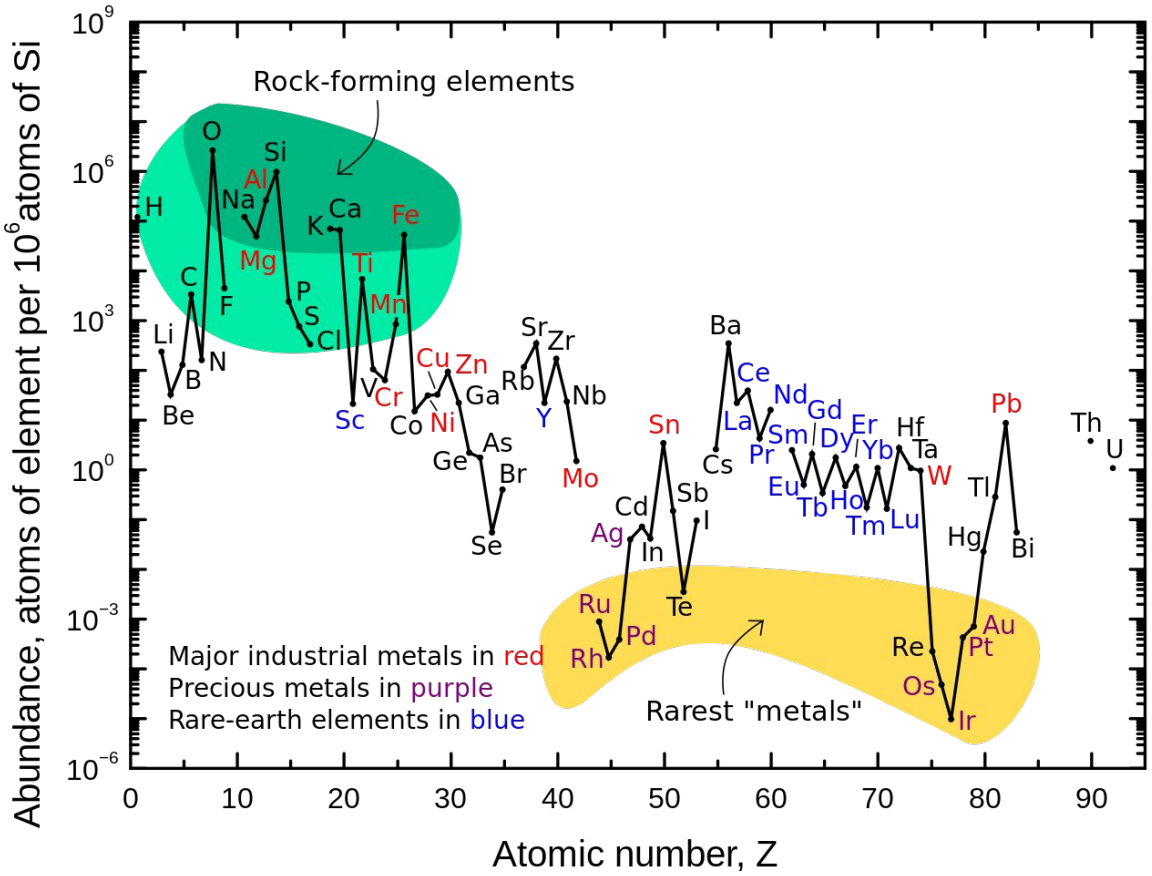
Table 9.1 Summary of the major stellar nuclear energy production phases and element synthesis

Fusing species	Main product	Byproducts	T (10^6 K)	Time scale (year)	Main reaction
H	He	^{14}N	35	8×10^6	$4p \rightarrow He$
He	C, O	^{18}O, ^{22}Ne	190	1.2×10^6	$3\alpha \rightarrow {}^{12}C$ $^{12}C + \alpha \rightarrow {}^{16}O$
C	Ne, Mg	Na	870	10^3	$^{12}C + {}^{12}C$
Ne	O, Mg	Al, P	1,600	0.6	$^{20}Ne + \gamma \rightarrow {}^{16}O + \alpha$ $^{20}Ne + \alpha \rightarrow {}^{24}Mg$
O	Si, S	Cl, Ar, K, Ca	2,000	1.3	$^{16}O + {}^{16}O$
Si	Fe	Ti, V, Cr, Mn, Co, Ni	3,300	0.03	$^{28}Si + \gamma \rightarrow {}^{24}Mg + \alpha$ $^{28}Si + \alpha \rightarrow {}^{32}S$
Fe	α		6,000	Collapse	$^{56}Fe + \gamma \rightarrow 13\alpha + 4n$

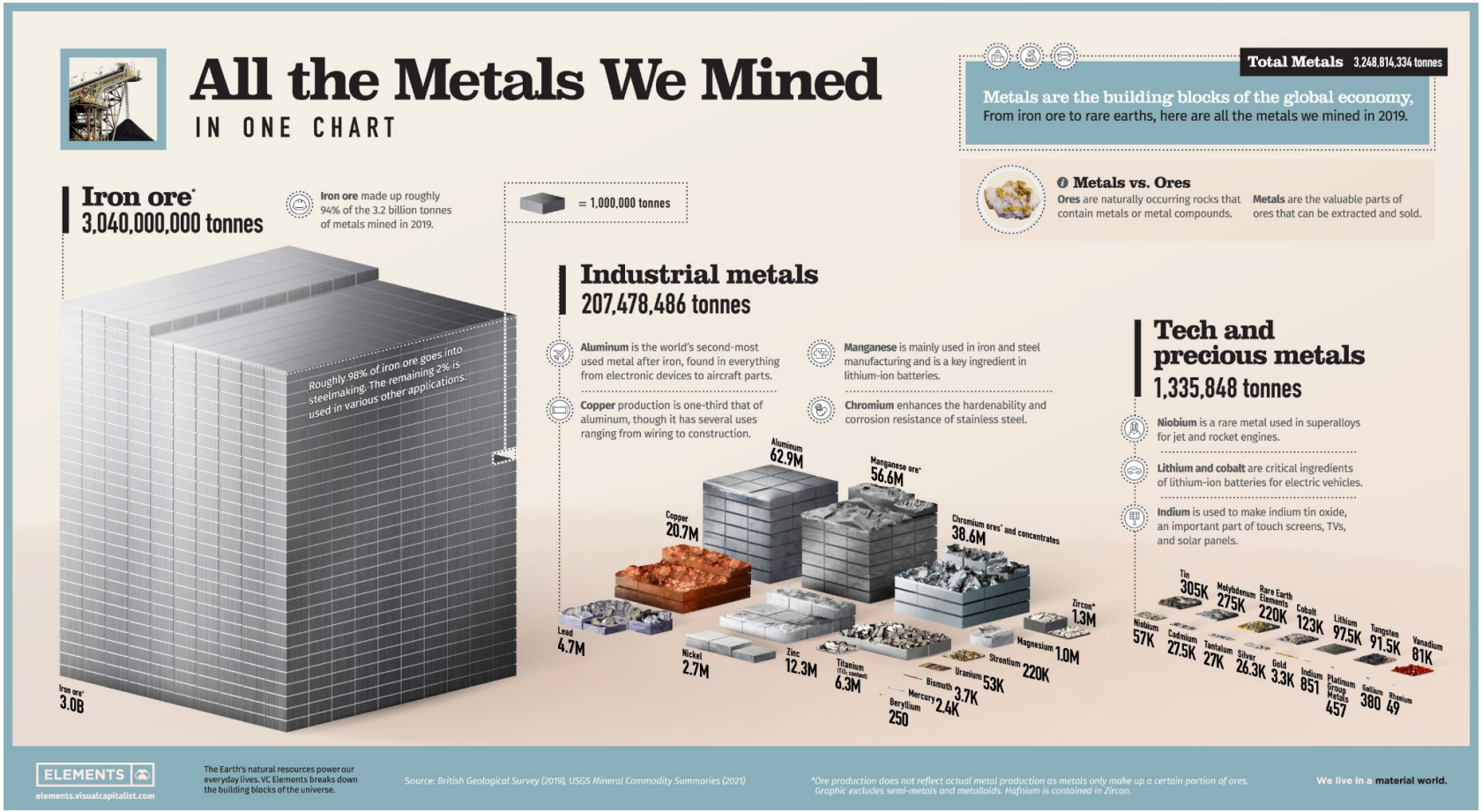
"We are all collections of chemicals made in the cataclysmic explosions of stars; we are stardust, or nuclear waste, depending on your perspective" - **Michael Brooks in *13 Things That Don't Make Sense***

G. Shaviv, The Synthesis of the Elements, Springer-Verlag 2012

Relative abundance of elements on earth's crust



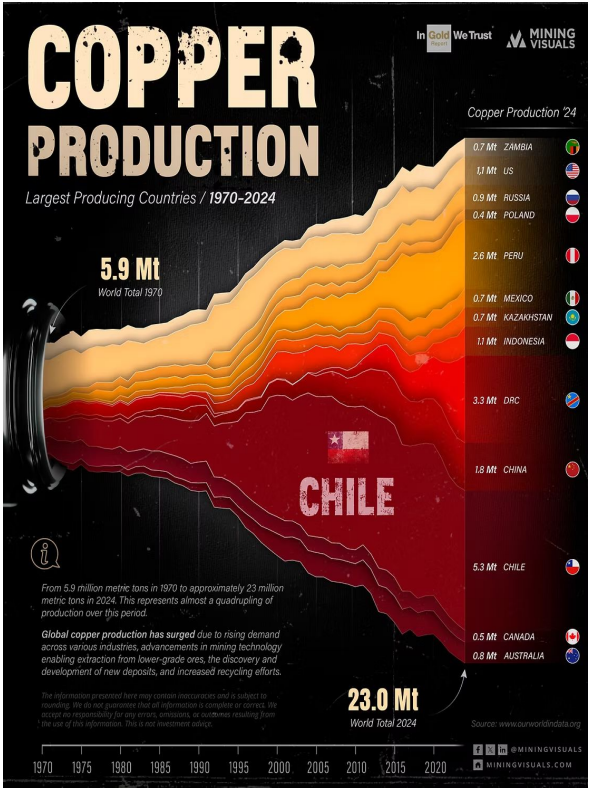
https://pubs.usgs.gov/fs/2002/fs087-02/

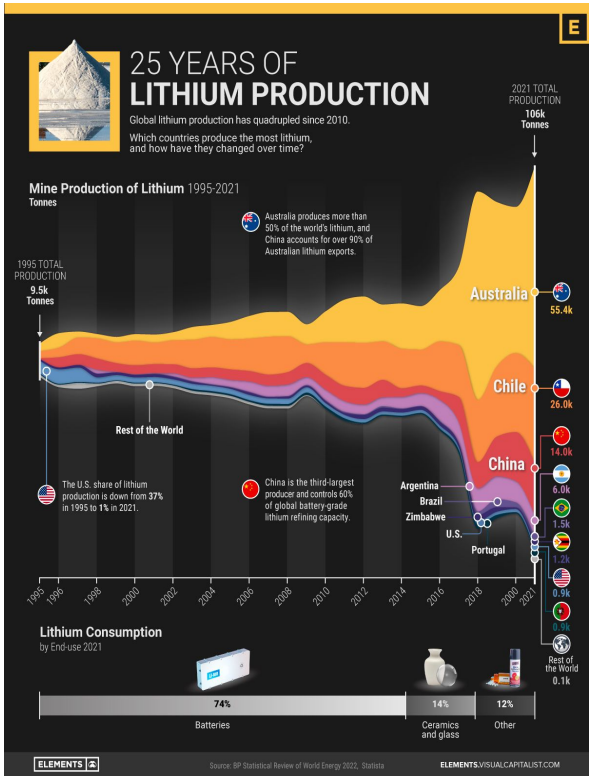


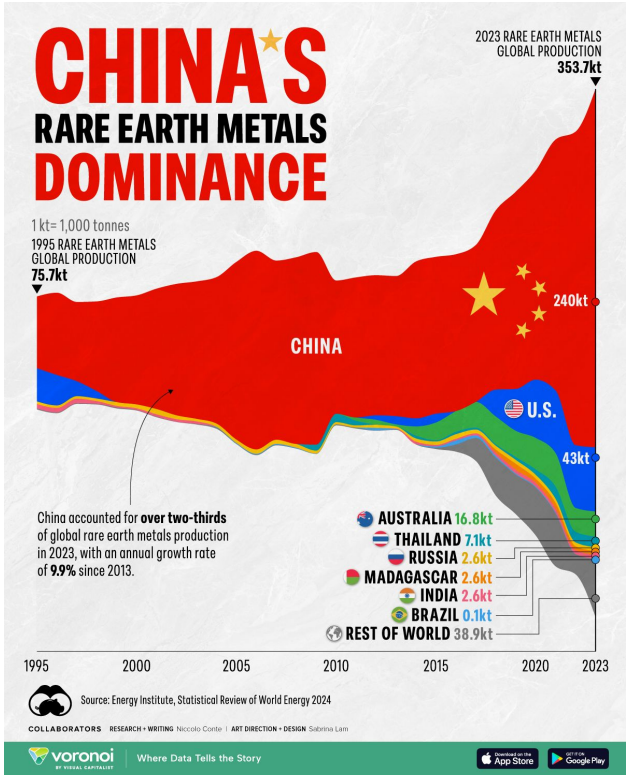
Identify three metals which are mostly (>50%) used for electrical/ electronic applications

https://elements.visualcapitalist.com/wp-content/uploads/2021/09/all-of-the-metals-one-visualization.html

Copper, Lithium and Rare earth metals

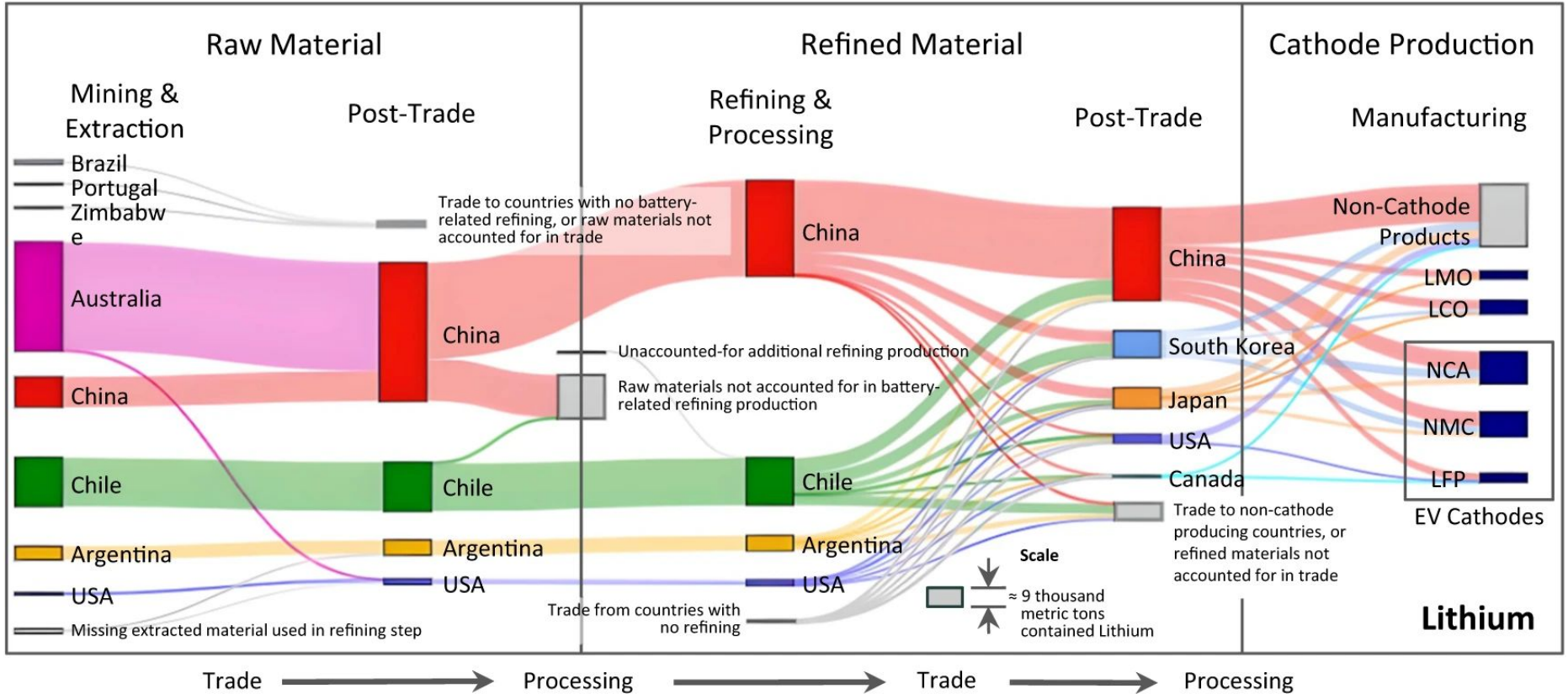






https://www.miningvisuals.com/post/charted-copper-production-1970-2024
https://www.visualcapitalist.com/visualizing-25-years-of-lithium-production-by-country/
https://www.visualcapitalist.com/visualizing-global-rare-earth-metals-production-1995-2023/

Sankey diagram (Material flow) of Lithium

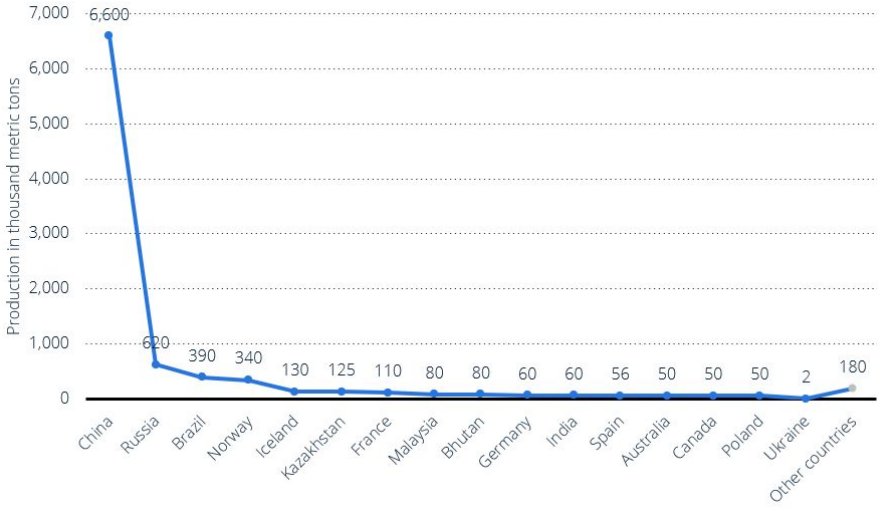


> 70% is used for batteries

https://doi.org/10.1038/s41467-024-46418-1

What about Silicon?

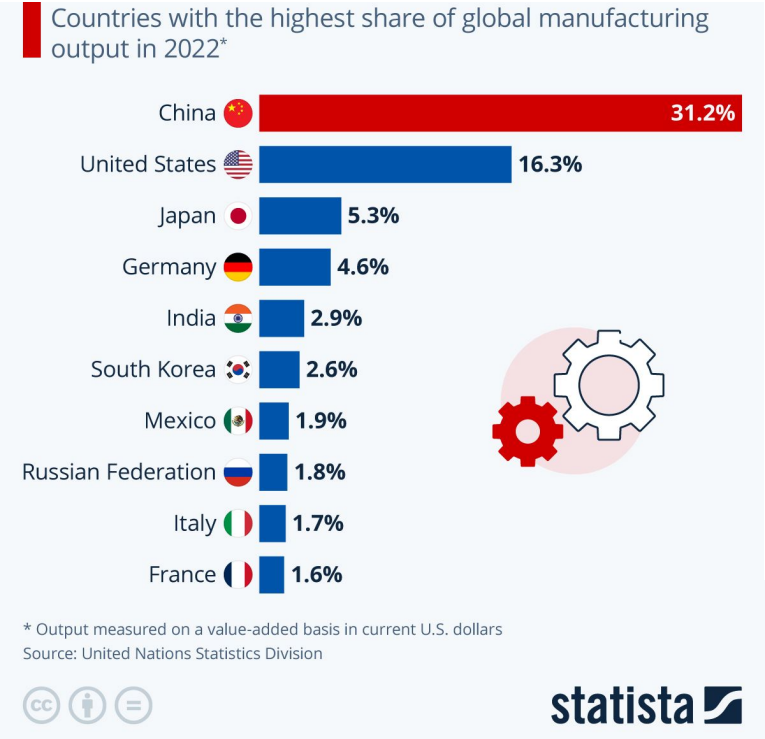
Silicon in the form of silicates constitutes more than 25% of the Earth's crust. Silica (SiO_2) as quartz or quartzite is used to produce silicon ferroalloys and silicon metal. Demand for silicon ferroalloys is driven principally by the production of cast iron and steel. Silicon is used primarily in the form of ferrosilicon for deoxidation and as an alloying agent in the production of iron and steel. Silicon metal, which generally is produced like ferrosilicon in submerged-arc electric furnaces, is used not as a ferroalloy, but rather for alloying with aluminum and for production of chemicals, especially silicones. Small quantities of silicon metal are processed into high-purity silicon for use in the semiconductor and PV industry.

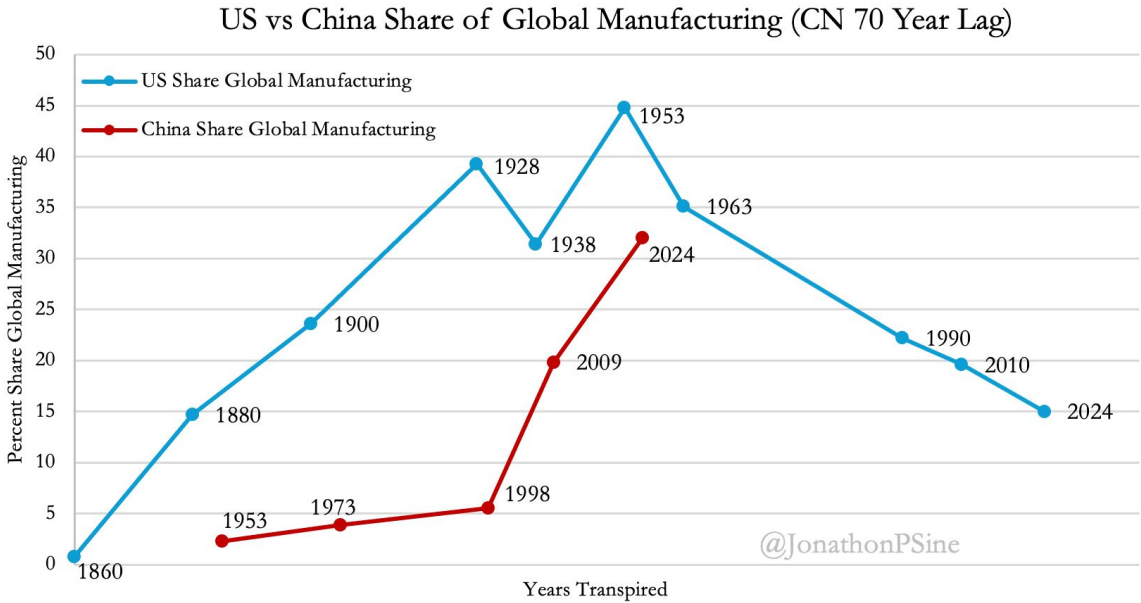


Global silicon production is ~9 million MT in 2023
China is the largest silicon producer
Polysilicon used for semiconductor and PV is < 10%

https://www.usgs.gov/centers/national-minerals-information-center/silicon-statistics-and-information
https://www.statista.com/statistics/268108/world-silicon-production-by-country/

Manufacturing capacity





U.S. attained 45% share of global manufacturing only after Europe's industrial capacity had been destroyed by WW2I

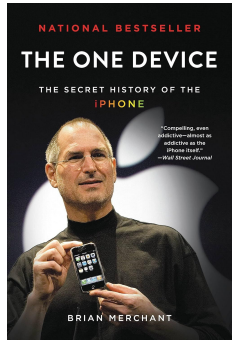
Corporations moved manufacturing where wages were lower, environmental and workplace regulation enforcement was lax

https://www.statista.com/chart/20858/top-10-countries-by-share-of-global-manufacturing-output/
https://www.cogitations.co/p/industrial-colossus-china-vs-1950s

Erik Loomis, Out of Sight:The Long and Disturbing Story of Corporations Outsourcing Catastrophe, The New Press 2015

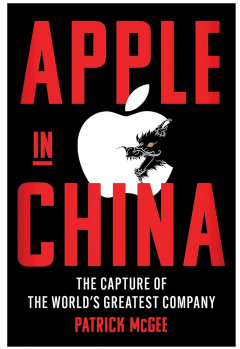
Why one shouldn't be clueless about Manufacturing

- Tech companies want to make gadgets which look and feel easy for the consumers to use.
- The complex and messy process of design and manufacturing is hidden from the consumer
- *There is no supplier out there where we (Apple) could just send a package of drawings and pack of specs and get what we wanted.*



- Product design - Features, performance

  Manufacturing - Cost, quality and reliability
- Understanding of manufacturing is needed to know why the real component looks very different from its schematic and textbook descriptions (eg: NOT gate)



- This also affects what kind of approximations one can take in component analysis
- How to improve the performance or reduce the size of the component/system

Why is Manufacturing important for nations?

A: Aircraft	1939	1940	1941	1942	1943	1944
Britain	7,940	15,049	20,094	23,672	26,263	26,461
United States	5,836	12,804	26,277	47,826	85,998	96,318
Soviet Union	10,382	10,565	15,735	25,436	34,900	40,300
Germany	8,295	10,247	11,776	15,409	24,807	39,807
Japan	4,467	4,768	5,088	8,861	16,693	28,180
Italy	1,750	3,257	3,503	2,821	2,024	-
B: Naval Vessels						
Britain	57	148	236	239	224	188
United States	-	-	544	1,854	2,654	2,247
Soviet Union	-	33	62	19	13	23
Germany (U-Boats only)	15	40	196	244	270	189
Japan	21	30	49	68	122	248
Italy	40	12	41	86	148	-
C: Tanks						
Britain	969	1,379	4,837	8,622	7,217	4,000
United States	-	331	4,052	24,997	29,497	17,565
Soviet Union	2,950	2,794	6,590	24,719	24,006	28,983
Germany (a)	794	1,651	3,298	4,317	5,993	8,941
(b)	-	394	944	1,758	5,941	10,749
Japan	559	1,023	1,216	1,271	891	371
Italy	(1940–43) 1,862 tanks, 645 self-propelled guns					
D: Artillery Pieces						
Britain	1,400	1,900	5,300	6,600	12,200	12,400
United States	-	c.1,800	29,615	72,658	67,544	33,558
Soviet Union	17,348	15,300	40,547	128,092	130,295	122,385
Germany	c.2,000	5,000	7,000	12,000	27,000	41,000

Richard Overy, Blood and Ruins, Penguin 2023

Manufacturing is important for nation's economy

a) As it brings foreign exchange through exports
b) Provides employment
c) Has a multiplier effect
d) Even services industry needs hardware in the cloud

and it also decides the outcome of a long war



https://www.politico.com/news/magazine/2023/06/09/america-weapons-china-00100373

Two ongoing material wars

Chip wars: how semiconductors became a flashpoint in the US-China relationship

The fight for the 21st century's most critical technology has become a major source of hostility between the world's two superpowers

The rules prohibit the sale, to Chinese customers, of **advanced chips** with both high performance (at least 300 trillion operations per second, or 300 teraops) and fast interconnect speed (generally, at least 600 gigabytes per second). Additionally, the rules restrict the sale of **fabrication equipment** if it will knowingly be used to make certain classes of advanced logic or memory chips. **The rules restrict not just U.S. companies, but citizens and permanent residents** as well. U.S. employees at Chinese semiconductor firms have had to pack up. ASML, a Dutch maker of fabrication equipment, has told U.S. employees to stop servicing Chinese customers.

https://spectrum.ieee.org/chip-ban#:~:text=The%20rules%20prohibit%20the%20sale,layer%20NAND%20chips%20last%20year.

THE WALL STREET JOURNAL.

BUSINESS | AUTOS

Automakers Race to Find Workaround to China's Stranglehold on Rare-Earth Magnets

Major manufacturers, fearful they will have to shut down assembly lines, consider moving some parts production to China

By *Sean McLain* Follow and *Ryan Felton* Follow

Updated June 4, 2025 11:23 am ET

https://the-ken.com/story/tvs-waits-tata-localises-mahindra-hedges-how-indian-automakers-are-dodging-a-magnet-meltdown

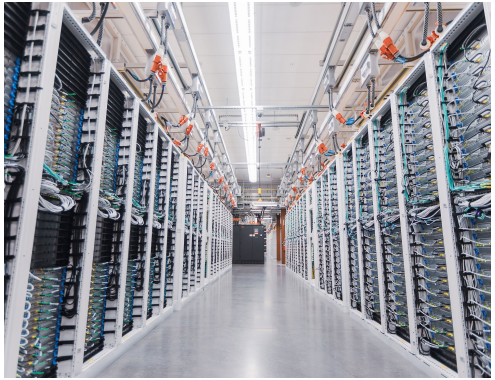
"In this time, the most precious substance in the universe is the spice melange… He who controls the spice controls the universe"

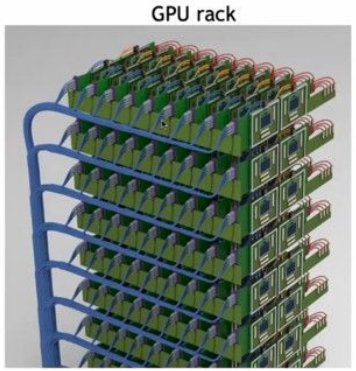
— Frank Herbert in Dune

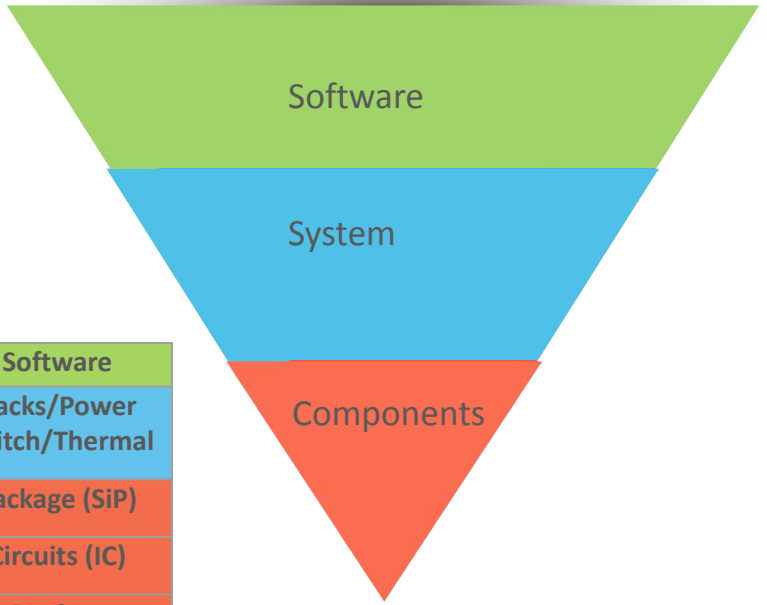
If the U.S. holds the upper hand in semiconductors, China has leverage over rare earths.

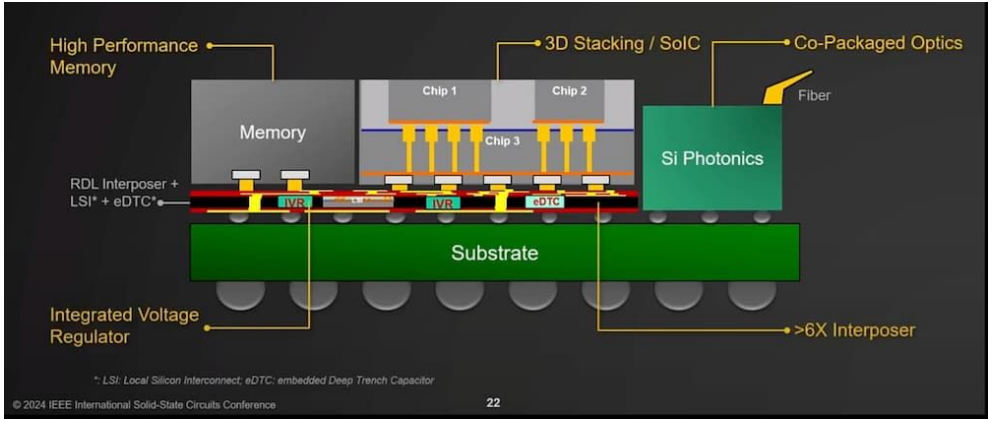
https://substack.com/home/post/p-166305576

A peak into a Modern-day Electronic System









Software
Racks/Power Switch/Thermal
Package (SiP)
Circuits (IC)
Devices
Structures
Materials

Kevin Zhang, ISSCC 2024
Kevin Zhang, 2025 TSMC Technology Symposium
https://spectrum.ieee.org/co-packaged-optics

What is covered in this course?

1. Materials science vs Condensed Matter Physics vs Solid State Physics
2. Importance of materials to modern engineering and EE in particular
3. Thermal expansion and conduction
4. Metals, Classical theory of electrical conduction: Drude model, Resistivity in pure metals
5. Limitations of classical theory of solids
6. Gibbs Free energy, Chemical potential, Partition function, Fermi-Dirac statistics

Quiz 1

7. Crystal structure, Miller indices, Reciprocal lattice and Brillouin zone
8. Basic quantum mechanics: Schrodinger's equation (3), Particle in a 1D infinite, finite well, Dirac well (3), quantum tunneling (1)

Quiz 2

9. Dirac-Kronig-Penney model (2), Energy bands in solids, E-k diagram, Holes (3), Density of states (2)
10. Intrinsic and extrinsic semiconductors (5)
11. Dielectric Properties in static fields – Polarization, Ferroelectric and Piezoelectric materials Properties in alternating field- Frequency Dependence, Dielectric loss, Equivalent Circuit (2)
12. Defects and its importance in material properties (2)
13. Phonons, Mechanical properties of solids (3)

EE2021 – Course logistics

Time table

- D slot for classes
- Every alternate Thursday slot is for tutorial.
- Tutorial problems will be released on Monday. You should come prepared

Evaluation

- Assignments (10%), Quiz1 (20%), Quiz2 (20%) & End sem (50%) on Insti dates
- Total = Assignment + Best of{Q1+Q2+Endsem, {Best of{Q1,Q2} +7/5*Endsem}}
- **No mobile use in classes**
- **Insti attendance rules (minimum 75% attendance to avoid a P grade)**
- **Use our EE mail address {sayak/deleep@ee} for any communication**

EE2021 – Teaching Assistants

ESB 106 (Lead TAs)	ESB 242 (Lead TAs)
Mansi Maheshwari EE24D026	**Nivedita Rai EE21D040**
Pranjal Singh Tomar EE24M057	**Praveena S EE24D002**
Vatsal Ramanuj EE22B159	Saiuditi EE22B139
Rishik Teja EE22B052	Sumedh Chatterjee EE22B005
Sundar Dhara EE24D302	Noel Manuel EP22b035
	Dweepee Sengupta EE24S059

EE and EP – Class reps

ESB 106	ESB 242
Sriram Venkata Krishna (EE24B067)	Daksh Gautam (EE24B099)
	Harshit Gautam (EE24B022)

On asking questions

- **The ability to ask clear and precise questions is one of the most important lifelong learning skills a student can acquire**
- A good question may reframe the problem for you to see it in an entirely new light and it also triggers more questions
- Improves your conceptual understanding
- Poor communication skills-a constant complaint from prospective employers

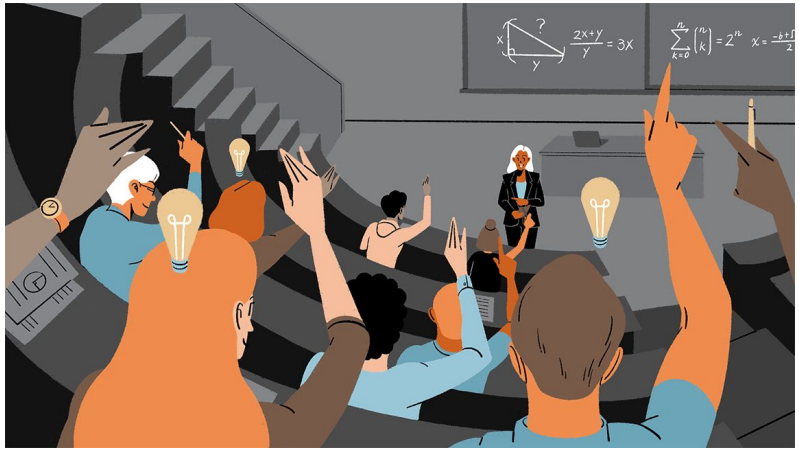
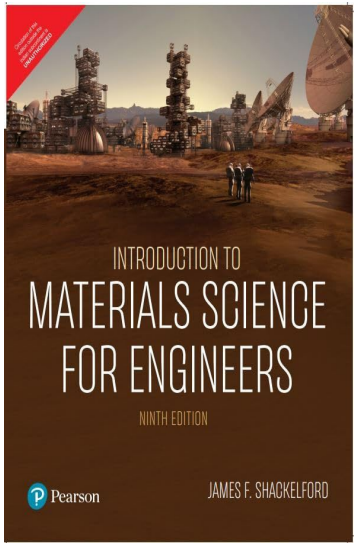
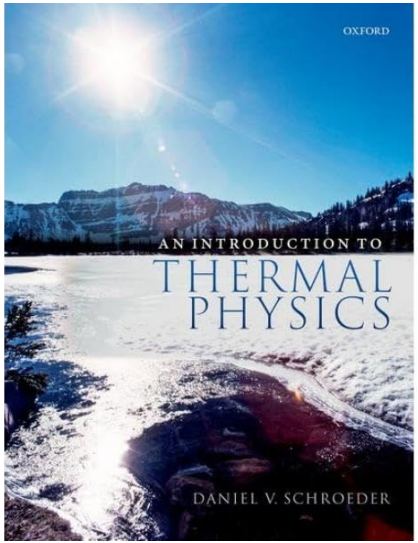


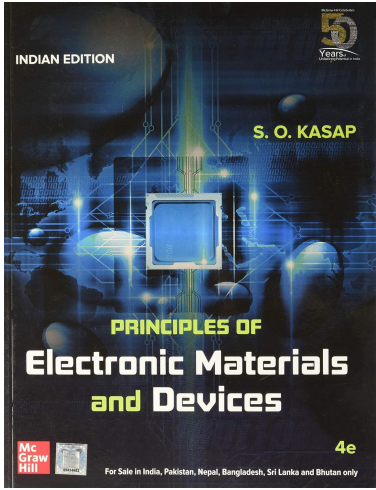
Image: https://www.chronicle.com/article/how-to-make-your-teaching-more-inclusive/

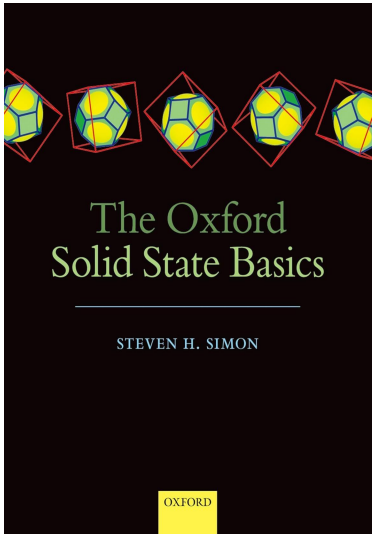
- **If you don't ask questions, your advisor/manager presumes that you have understood everything**

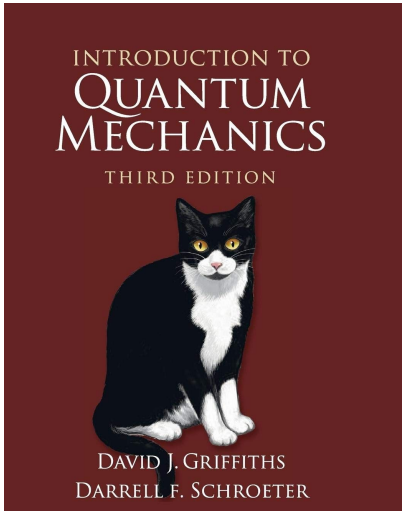
Textbooks


INTRODUCTION TO
MATERIALS SCIENCE
FOR ENGINEERS
NINTH EDITION
Pearson
JAMES F. SHACKELFORD

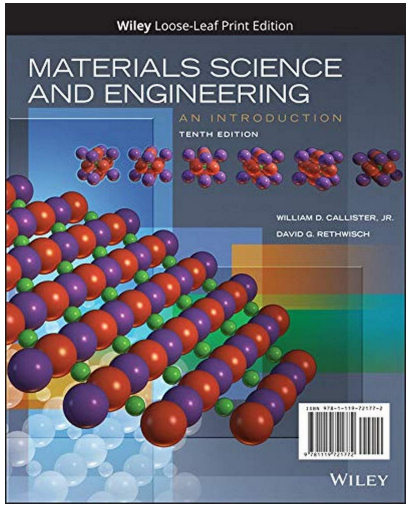

OXFORD
AN INTRODUCTION TO
THERMAL
PHYSICS
DANIEL V. SCHROEDER

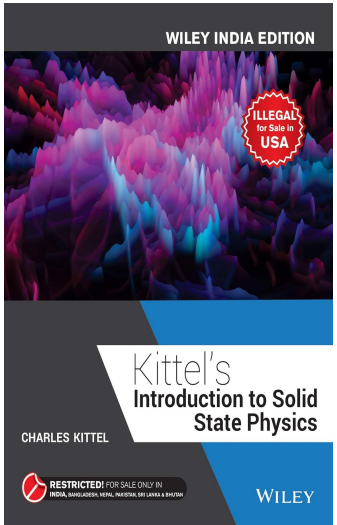

INDIAN EDITION
S. O. KASAP
PRINCIPLES OF
Electronic Materials
and Devices
4e
For Sale in India, Pakistan, Nepal, Bangladesh, Sri Lanka and Bhutan only

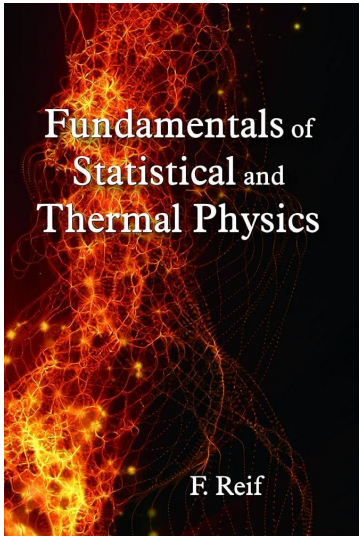

The Oxford
Solid State Basics
STEVEN H. SIMON
OXFORD

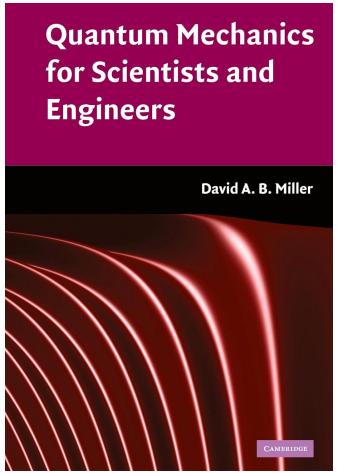

INTRODUCTION TO
QUANTUM
MECHANICS
THIRD EDITION
DAVID J. GRIFFITHS
DARRELL F. SCHROETER

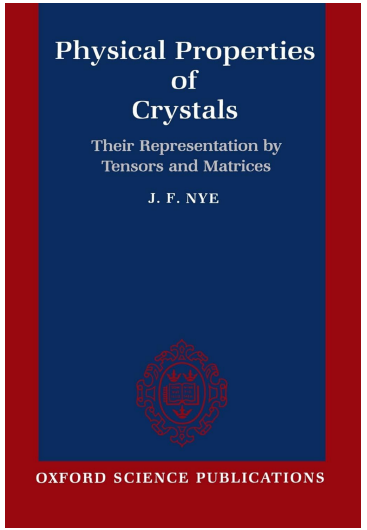
Reference books









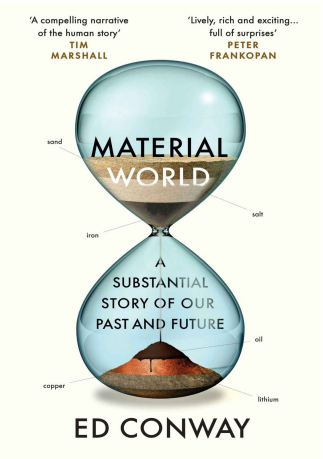


Popular science books

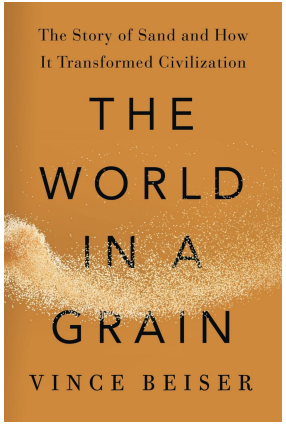
1. Material World by Ed Conway
2. Materials and Dematerialization by Vaclav Smil
3. The One Device: The Secret History of the iPhone by Brian Merchant
4. Volt Rush by Henry Sanderson
5. The War Below by Ernest Scheyder
6. The New Science of Strong Materials by J. E. Gordon
7. Why Things Break by Mark Eberhart
8. The World in a Grain by Vince Beiser
9. Chip War by Chris Miller
10. Apple in China by Patrick McGee

Assignment #1

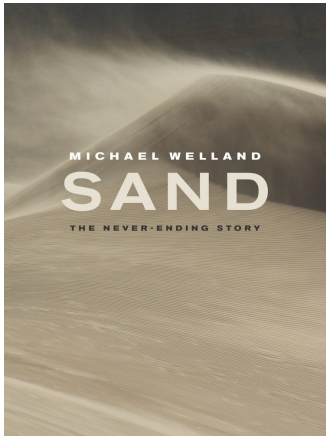
An entrepreneur has come up with an idea to make Silicon wafers in the vicinity of Thar desert. They have approached Union govt for 50% subsidies and State govt of Rajasthan for 20-30% subsidies under India Semiconductor Mission to setup a plant to make Silicon wafers out of sand. They claim that these wafers can be used by the semiconductor fabrication facilities coming up in the neighbouring state of Gujarat. They claim that this will help in creating a truly domestic supply chain and support the efforts to make India Atma Nirbhar and Viksit by 2047. Both govts have approached you to check the veracity of their claims and the feasibility of the proposal. Read the relevant chapter of these books and concise 1 page evaluation report (**pdf**) by 10th August (Sunday 11:59 PM)citing **only scientific and engineering reasons**.



Chapter- The Longest Journey



Chapter- High Tech, High Purity



Chapter- Servant: Sand in our lives